6DoF navigation in virtual worlds: comparison of joystick-based and head-controlled paradigms

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ABSTRACT

6DoF navigation in a virtual world can usually be implemented by two types of navigation techniques: joystick-based input devices and steering metaphors based on movements of the user's body, e.g. head-controlled paradigms. These two different types of 6DoF navigation techniques provide users with the same level of user control, but the latter introduces the user's physical movements in the navigation, which we believe will improve the navigation experience in immersive virtual environments. In this paper, we compare these two types of 6DoF navigation techniques in an immersive context; through an experiment using both objective and subjective measurements to measure user performance, the occurrence of cybersickness symptoms and the level of presence, when using either of these navigation paradigms.

Categories and Subject Descriptors

I.3.7 [**Computer graphics**]: Three-Dimensional Graphics and Realism – *Virtual Reality.*

General Terms

Performance, Experimentation, Human Factors

Keywords

Virtual Environment, 6DoF Navigation, Cybersickness, Presence.

1. INTRODUCTION

Navigation is one of the essential tasks for a user to explore virtual or augmented worlds. This is the case with any immersive system, from HMD devices to CAVE-like environments, when we want to explore a world that is larger than the physical limit of the setup. Several techniques have been proposed to perform navigation according to the nature of the task, such as flying or walking. It can be pointed out than most of these paradigms were designed in order to mimic a realistic behaviour of a flying plane, a space ship, a car, or a natural human walking (e.g. walking-in-place [9], redirected walking [21], walking simulation devices [4]…).

Depending on the scenario of the virtual experience, navigation can be controlled by specific input devices, such as a steering wheel for driving a vehicle, or a joystick or haptic device when physical involvement is required or when haptic manipulation tasks must be performed [6]. When the user needs both hands free to manipulate objects, head tracking [22] or body tracking devices are well-adapted methods to control the navigation.

In most cases, it is not necessary to have all the degrees of freedom for navigation; it may even be preferable to reduce the number of degrees of freedom based on the nature of the content of the virtual scene which is to be explored, in order to achieve a more intuitive and more efficient navigation (e.g. walking on the ground). However, in some other applications, it is interesting to use 6 degrees of freedom when virtual contents do not exert any constraints on the user, e.g. when flying in outer space or diving underwater, or exploring scientific data in immersive context.

In this paper we compare two 6DoF navigation techniques with which the user has exactly the same level of control, but different levels of physical involvement in the task. In both cases, visual immersion is provided by a CAVE-like system, and users wear a pair of tracked shutter glasses, in order to have a correct stereoscopic view of the virtual scene when navigating. For the first navigation technique, the user is equipped with a joystick-based device (see Figure 1) to manage 6DoF inputs. For the second technique, 6DoF navigation is controlled by the position/orientation of the user's head relative to a neutral referential.

To compare these two techniques, we have measured user performance, as well as users' feeling of presence and their level of cybersickness. One of our main goals is to study whether the user physical involvement will make a difference in terms of user experience, especially in term of cybersickness.

The paper is organized as follows. In section 2, we discuss some existing research related to 6DoF navigation, physical involvement for navigation and studies about different factors to evaluate this navigation. Section 3 presents the technical features

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and procedure of the experiment that we carried out. Section 4 describes the results of the experiment. Finally, in section, 5 we discuss these results and in the last section, we conclude the paper and describe some possibilities for future work.

2. RELATED WORKS

2.1 6DoF navigation techniques

In Virtual Reality systems such as CAVEs or HMDs, a mouse, keyboard or any other input device can be used to perform navigation in the virtual scene without moving in the real world. However, these are generally not a good choice since a more natural navigation paradigm is required in order (a) to improve the user's level of immersion/presence in the VR system; and (b) to facilitate the interactions between the user and the system (e.g. hands-free navigation [12]).

Slater et al. have shown that natural walking can provide users with the best level of presence because in this case, there are no conflicts between the user's visual system and vestibular/ proprioceptive systems [17]. However, it's not always possible to use natural walking motion to navigate a virtual scene, because of the limited size of the physical workspace. Therefore, several navigation paradigms have been developed to tackle this problem, such as walking-in-place [9], redirected walking [21], walking simulation devices [4], and so on).

However, the walking metaphor is far from sufficient for all kinds of virtual contexts. Sometimes the flying metaphor is required if we want to fly through the virtual scene. This is especially true for the immersive exploration of non-realistic virtual scenes, for instance, such as analyzing 3D data or exploring numerical simulation results in different scientific application fields. In this case, 6DoF navigation paradigms are more appropriate to accomplish the task.

6DoF navigation can be supported by directly mapping the user's viewpoint with various input devices like a flystick, 3D mouse/space mouse, joystick and even haptic arms [5]. Besides these input devices, 6DoF navigation can also be performed by relying on the movements of the user's body. Such systems can be summarized as steering metaphors [2]. The steering metaphor is generally easy to understand and provides the highest level of control to the user. For example, Ware and Osborne proposed the Camera-in-Hand technique. The work area of a 6DoF tracker fixed on the user's hand defines a miniature of the scene to control a virtual camera [24]. Bourdot and Touraine [1] proposed an approach where the virtual navigation is controlled by applying damping on the actual 6DoF signal of a tracker with respect to a neutral referential. This technique may be applied to hand tracking, but has mainly been designed for hand-free navigation via head tracking.

The steering metaphor provides different proprioceptive feedback for the user than the walking metaphor. Nevertheless, the vestibular and proprioceptive systems of the user are engaged during the navigation process; we believe this will make a difference in terms of user experience compared with 6DoF navigation using input devices like joystick or flystick, etc.

2.2 Evaluation of navigation techniques

It's not obvious to compare different navigation techniques without a given virtual context and the associated task. Nevertheless, Bowman et al. [3] have proposed a list of quality factors which can help us to compare and measure the performance of navigation techniques including general performance indicators such as velocity and accuracy, spatial awareness, ease of learning, ease of use, and level of presence.

General performance indicators are relatively easy to measure. We can choose factors that we need depending on the given task and the configuration of the virtual scene, like the time required to complete a navigation task, the accuracy of actions (e.g. proximity to the target if we ask the user to get to an object), length of the trajectory and number of backtracks, etc.

The level of presence is an important factor for the use of navigation techniques in an immersive virtual environment. Various research has been done on the nature of presence in virtual environments, and several definitions and theories have been proposed. Slater et al. [16] proposed an important distinction between "presence" and "immersion": presence is a subjective phenomenon that can be defined as the sensation of being in a virtual environment, while immersion is an objective description of aspects of the system. Witmer and Singer [25] relate presence in part to the concept of attention: presence may vary across a range of values that depends partly on the allocation of attentional resources. These authors believe that both involvement and immersion are necessary for experiencing presence.

These definitions do not necessarily contradict one another, but they can have different implications. As a result, the way that presence is measured depends on which theory is used. Among all the approaches that are proposed, the most commonly used measures are based on subjective ratings through the use of questionnaires. Slater et al. have proposed a questionnaire [23] based on the feeling of being present in the virtual environment while Witmer and Singer have constructed another questionnaire based on the involvement of attention [25]. Schuemie et al. [15] have proposed a taxonomy of different measures of presence and a detailed analysis of factors that influence presence in the virtual environment.

Another important factor that we should take into consideration when evaluating a navigation paradigm is user comfort. In Virtual Reality studies, particularly those related to navigation tasks, one key aspect of comfort is the lack of symptoms related to cybersickness.

Ergonomic studies have tried to identify perceptive factors that are susceptible to cause cybersickness when using a virtual environment. Rich and Braun [14] have examined the relationship between the user's level of control of his/her own movements and cybersickness, while So et al. [18] have carried out experiments to study the influence of the navigation speed (both linear and circular components) on the level of severity of cybersickness. So, Lo and Ho [19] have emphasized that the visual complexity of the virtual scene should also be taken into consideration in order to characterize the relationship between a virtual scene and the occurrence of cybersickness. Another study by Stanney et al. [20] gives us a better global view that addresses all the perceptive factors that we have mentioned above.

To evaluate the level of cybersickness, the Simulator Sickness Questionnaire or SSQ [10] remains the reference tool despite the development of various physiological measurements. Kennedy et al. have identified 3 main factors and each one can be associated with several symptoms: Nausea (nausea, stomach awareness, etc.), Oculomotor (eyestrain, blurred vision, etc.) and Disorientation

(dizziness, vertigo). LaViola has also discussed about the physiological aspects and theories of cybersickness [11].

3. METHODS

3.1 Participants

Twenty participants took part in the experiment (12 male, 8 female) with ages ranging from 18 to 49 years old (mean=28 years). All participants were experienced in the use of VR systems.

3.2 Materials

The experiment was carried out within a large CAVE-like immersive environment with a volume of approximately 60 cubic meters using four retro-projection screens surrounding the user (the floor and three walls).

Participants are equipped with a pair of shutter glasses in order to see stereoscopic images, and a tracking system with infrared cameras is installed to record spatial information regarding the position of the users

On the one hand, for the joystick-based navigation, the joystick used in the experiment is a Logitech cordless Rumblepad that is configured as in Figure 1, so as to support 6DoF input. We used the left stick and two buttons on the left side to map rotational controls (marked in blue), and the right stick and two buttons on the right side to map translational controls (marked in green). In this way, users can hold the joystick comfortably without looking for buttons during the navigation (thumbs for the two sticks, two fingers for each pair of two buttons). The sticks and buttons provide acceleration for the virtual viewpoint instead of giving a constant velocity: the longer the button is pushed, the greater the navigation speed. This is intended to let the user have full control of the navigation speed.



Figure 1: Joystick configuration for 6DoF navigation.

On the other hand, for the head-controlled navigation paradigm, the steering metaphor we use is based on the user's body motion via 6DoF head tracking (see Figure 2 and Figure 3). Before the navigation begins, the user chooses a neutral position and direction during the calibration phase (normally the user is situated in the central part of the CAVE and faces the front screen). Following this, the user can move relatively to the neutral position to achieve horizontal translations in the virtual scene (e.g. step forward to go forward in the virtual world), but can also stand on the toes to go upwards or bend the knees to go downwards. This translational navigation continues as long as the user does not return to the neutral position. For rotations, the user turns his/her head and the viewpoint in the virtual world will follow. In this way, when the user is rotating towards a point of interest, the virtual scene is rotating in the opposite direction, in such a way that when the user comes back to his/her initial neutral direction, the rotational navigation will stop "naturally". The speed of translation/rotation is proportional to the user's amplitude of motion.

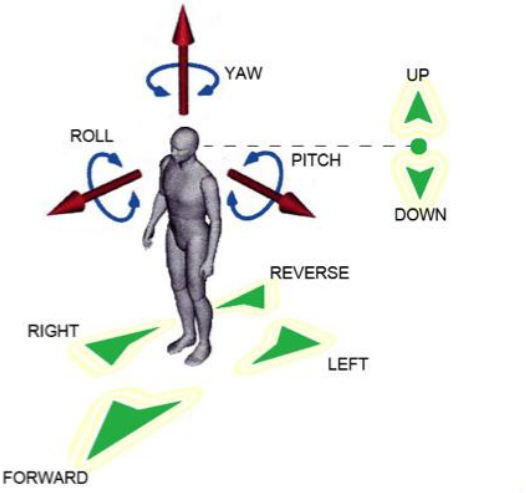


Figure 2: Head-controlled navigation paradigm.

We chose this head-controlled steering metaphor because it gathers a number of interesting features, such as: 6DoF control of the virtual navigation, vestibular stimulation of the user, and no additional device is required, because head tracking is necessary for head-tracked stereoscopy in a CAVE device.



Figure 3: 6DoF navigation using head-controlled technique (the user turns his/her head to perform rotational roll movements)

3.3 Procedure

The participants were invited to take part in a user evaluation of VR interaction devices: the joystick condition (named J) and the head-controlled condition (named H).

3.3.1 Pre-session questionnaires

The participants began by filling a short questionnaire measuring susceptibility to motion sickness [7] and the overall level of malaise [13]. Then each participant completed a copy of the Simulator Sickness Questionnaire (SSQ) (as mentioned in part 2.2), a standard questionnaire used to measure simulator sickness to record the initial sickness level of the participant.

3.3.2 Training stage

Having completed the questionnaires, each participant spent about 3 minutes to get familiar with the navigation technique for each condition. We explained to participants how the navigation technique worked, and let them try to achieve a simplified version of the task used in the real test.

3.3.3 Experimental stage

In the test, participants carried out a navigation task within a 3D virtual slalom course. The task took place within the virtual replica of a canyon, in which ten circular gates were placed and numbered 1 through 10 (see Figure 4).

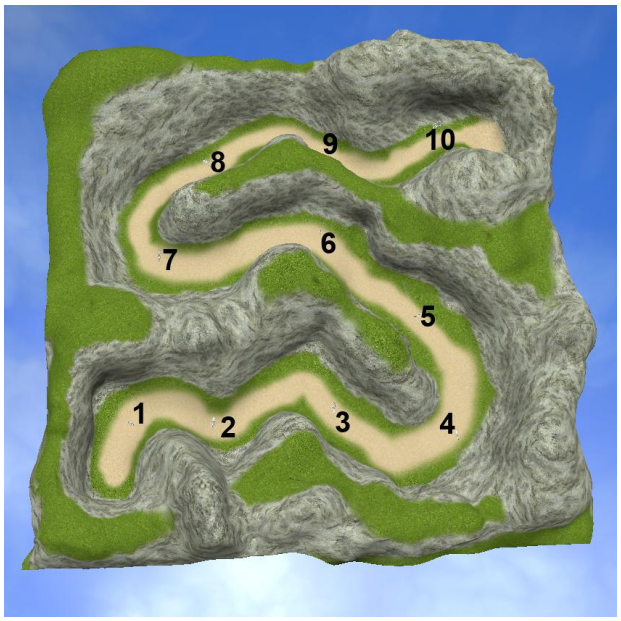


Figure 4: Top-view of the virtual slalom.

Each gate included a colored section along part of its perimeter. This colored section contains two colors that are used to give users appropriate indications of the right direction for the navigation. The green color marks the front side of the gate and the red color marks the back side (see Figure 5).

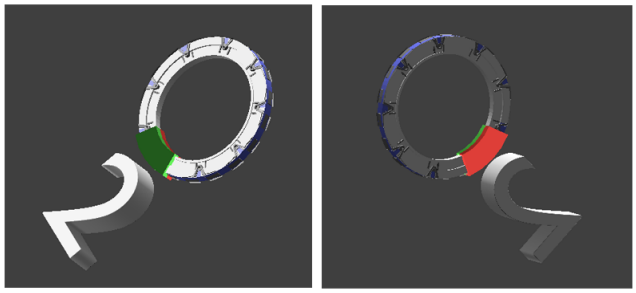


Figure 5: Front side of the gate (left). Back side of the gate (right).

Participants were instructed to complete the course as quickly as possible, passing through each of the ten gates by the front side in ascending numerical order, making sure that the green coloured section was beneath them (see Figure 6). This allowed us to ensure that the course required truly three-dimensional navigation.

Each participant completed the course twice, one for the J condition, the other for the H condition. A counterbalanced design was used to control the effects of condition order.

3.3.4 Post-session questionnaire

After completing each condition, participants completed a 42-item posttest questionnaire including 16 items from the Simulator Sickness Questionnaire (SSQ), and 26 items from Witmer and Singer's [25] presence questionnaire (PQ). Witmer and Singer's original questionnaire comprises 32 items, but not all of these items were applicable to the situation studied here.

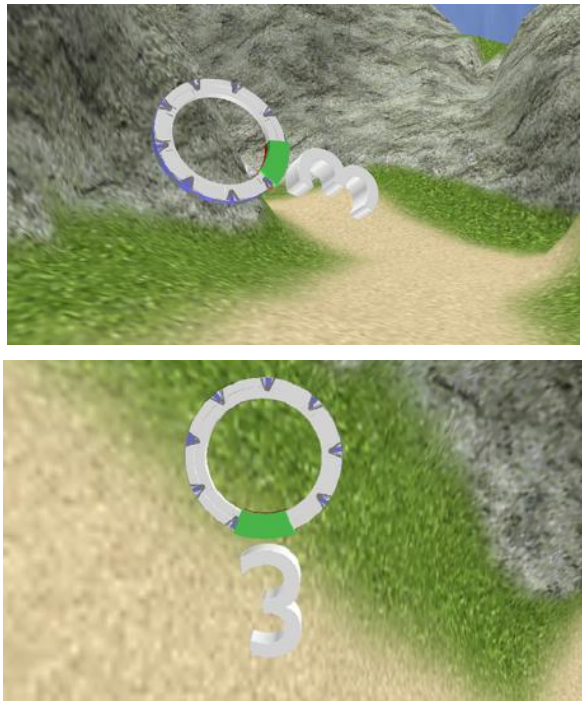


Figure 6: Participants should get through the gates by the front side (green instead of red) with the colored section underneath as indicated in (bottom).

4. RESULTS AND ANALYSIS

4.1 Dependent variables

In addition to questionnaire responses, which constitute a source of subjective data related to the user experience in this virtual navigation task, we gathered a number of objective performance measurements throughout the course:

- Course completion time: the time that the participant took to complete the course (in seconds).
- Curvilinear distance: the total length of the 3D path of each participant (in meters).
- Angular distance: the total rotational movements of the participant throughout the course (in degrees). Angular distance is computed and considered as an index of the extent to which user navigation was straightforward (low angles) or, conversely, erratic (high angles).
- Linear velocity: the average speed of translational movements of the navigation (in meters per second, m/s).
- Angular velocity: the average speed of rotational movements of the navigation (in degrees per second, degree/s).

As final measures of performance, we examined the position of the user when passing through each of the gates and collected:

- The sum of distances of the user to the center of the gates from No. 1 to 10 (measured in meters).
- The sum of angles of the user's position relative to the center of the green strip of the gates from No. 1 to 10 (measured in degrees). We projected the user's upper body orientation onto the gate's plane and calculated the 2D angle between the projected line and the diameter of the gate that crosses the centre of the green strip.

These two variables were intended to measure whether users were able to adopt a correct orientation during the navigation task.

4.2 Objective data analysis

A paired-samples t test was carried out using JMP 9 on course completion time (see Figure 7 and Figure 8), curvilinear distance and angular distance, linear velocity and angular velocity (see Figure 9), the distance to the centre of the gates and angle to the middle of the green strip, using the navigation paradigm as a within-subjects variable.

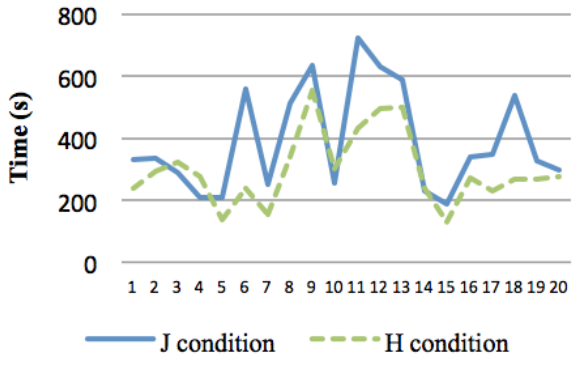


Figure 7: Distribution of the course completion time for the J condition (blue) and H condition (green)

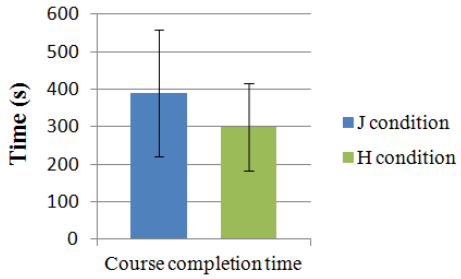


Figure 8: Average value and standard deviation completion time for J condition (blue) and H condition (green).

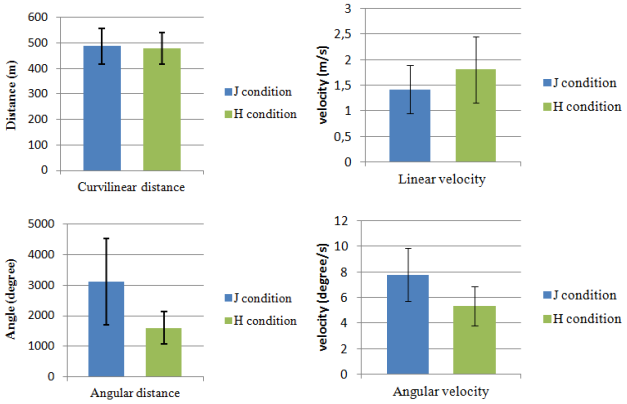


Figure 9: Average value and standard deviation of curvilinear distance (top left), angular distance (bottom left), linear velocity (top right) and angular velocity (bottom right) of the 3D path for J condition and H condition.

The paradigm exerted a significant effect on five of these variables ($p<0.005$): on average, participants completed the course faster (M= 298 vs. 390 s), travelled at a greater linear velocity (M=1.80 vs. 1.42 m/s), chose a more direct, i.e. shorter route (M=478.4 vs. 487.2 m) and performed fewer rotational movements in terms of angular distance (M=1607 vs. 3107 degrees) in the head-control navigation condition compared with the joystick-based navigation condition. In addition, average angular velocity was significantly lower in head-controlled navigation versus joystick-based navigation (5.31 vs. 7.77 degree/s), suggesting a finer control of rotational movements in the latter condition.

Effect sizes were also calculated using Cohen's *d* index. The effect of the navigation paradigm on the angular distance (d=1.15) was large. Its effect on course completion time (d=0.60) was medium-sized. The effects of the paradigm on the curvilinear distance, however, were small (d=0.13). Finally, the effect of the navigation paradigm on the distance to the centre of the gates and angle to the middle of the green strip were not significant.

These results suggest that virtual navigation using the head-controlled technique is much less erratic and reasonably quicker than when using a joystick to navigate the scene. However, the distance travelled was only slightly greater when using the joystick. One way to interpret these findings is to say that head-controlled navigation allows users to gain some time by making it

easy for them to position themselves more accurately with respect to their environment.

4.3 Subjective data analysis

4.3.1 Simulator sickness

SSQ results were interpreted following the guidelines proposed by Kennedy et al. [10]. Participant responses yielded scores for each of the three components of simulator sickness proposed by these authors – Nausea (N), Oculomotor (O) and Discomfort (D) – as well as a total score (T).

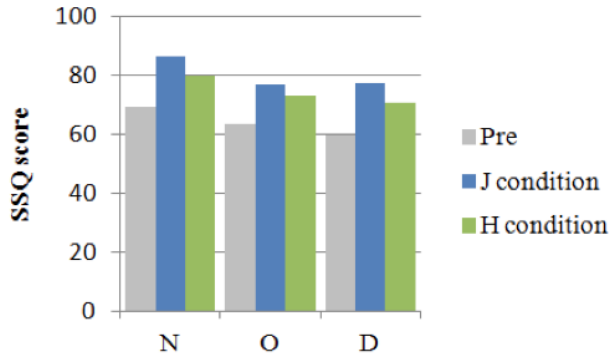


Figure 10: Average value of the three components of SSQ questionnaire for the pre-session test (grey), the J condition (blue) and H condition (green).

A paired-samples t test was carried out on each of these variables using the condition as a within-subjects variable. SSQ scores rose significantly in both conditions compared with pre-exposure scores ($p<0.005$). The navigation paradigm exerted significant effects on the total sickness score and on the nausea scores ($p<0.002$): participants experienced significantly less symptoms of nausea and general cybersickness in the H condition versus J condition (see Figure 10 and Figure 11).

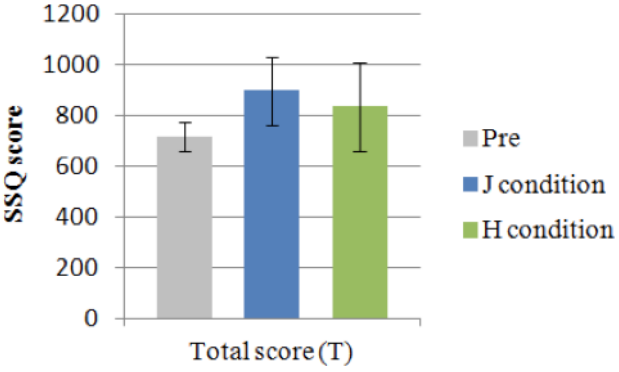


Figure 11: Average value and standard deviation of the total score (T) of SSQ questionnaire for the pre-session test (grey), the J condition (blue) and H condition (green)

The effect of the navigation paradigm on the total cybersickness score (d=0.41) and on the nausea score (d=0.40) was medium-sized.

4.3.2 Presence

Results to the presence questionnaire (see Figure 12) were interpreted using the guidelines outlined in Witmer and Singer [26]. A paired-samples t test was carried out on the total score, using the condition as a within-subjects variable. The navigation paradigm exerted a marginally significant effect on the level of presence ($p=0.006$). Participants experienced a greater level of presence in head-controlled navigation than in joystick-based navigation (M=121.3 vs. 114.2). As before, Cohen's d was computed, yielding a medium effect size (d=0.54).

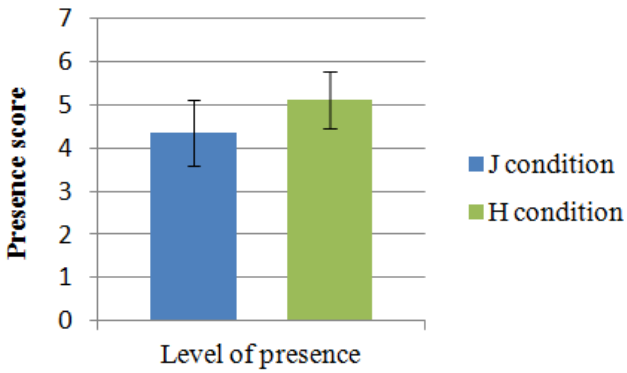


Figure 12: Average value and standard deviation of the presence questionnaire for the J condition (blue) and H condition (green).

5. DISCUSSION

The objective and subjective results suggest a number of benefits related to the use of head-controlled rather than joystick-based navigation. The use of the head-controlled paradigm increased efficiency of navigation in the virtual scene, both in terms of completion time and of trajectory quality.. Participants were able to exert a far greater control over their trajectory, leading to (a) a more rapid completion of the navigation task and (b) far more direct, less erratic travelling from one gate to the next in a 3D space.

Concerning the trajectory quality, both curvilinear distance and angular distance are significant. Although this effect is small for curvilinear distance, we highlight the fact that the effect is large for angular distance. That means that in the joystick condition, users need to adjust their orientation more often to pass thougth the next gate with the expected orientation. Indeed, participants using joystick-based navigation found themselves to be constantly in need of adjusting their own position and orientation, to locate the successive gates and to go through them correctly. An explanation might be that it is more difficult for subject to simultaneously control orientation and position when using the 6DoF joystick navigation technique, rather than when using the head-controlled paradigm.

In contrast, head-controlled navigation allowed users to adopt the correct orientation more easily. Crucially, this was achieved by involving their entire body to control movement, rather than just their hands. When using a joystick, users often found it difficult to adopt a correct orientation during the 6DoF navigation task. This informal feedback corroborates the results derived from analyzing objective performance data. Informal user feedback also taught us that the head-controlled navigation paradigm was in most cases preferred to joystick-based navigation.

Furthermore, head-controlled navigation led to a decreased severity of cybersickness and a moderate increase in the feeling of presence during this navigation task. These positive effects of head-controlled navigation – over joystick-based navigation – on user performance, wellbeing, and presence, could be explained by arguing that head-controlled navigation leads to an *increased physical and psychological involvement* of the user in the virtual environment. Based on the data presented in this paper, this construct might be discussed in three different ways.

First, current knowledge on the aetiology of cybersickness rests on two key models [11]. On the one hand, cybersickness derives from a conflict between sensory inputs to the central nervous system. On the other, it has also been claimed to have links with disruptions in the control of posture. Both of these models might explain the reduced occurrence of cybersickness when using head-controlled navigation. Indeed, head-controlled navigation paradigm the user with a direct connection between visual perception of the virtual scene and proprioceptive feedback from physical motion in the real world. One explanation for our results would be that this proprioceptive feedback is far more present in head-controlled rather than joystick-based navigation, leading to consistency between the visual and proprioceptive channels and to increased postural stability.

Second, the use of the head-controlled navigation paradigm was also related to a higher sense of presence, which has been likened to a form of psychological involvement of the user. The connections between the physiological and psychological mechanisms of presence in a virtual environment are not well known today. When using joystick-based navigation, participants made a lot of efforts to complete the course with a satisfactory level of performance and to comply with the instructions related to how they should pass through the fates. The results regarding course completion times and angular distance suggested that they felt the need to constantly adjust their trajectory to compensate for the limitations of joystick-based navigation when using this paradigm. We believe that this may explain why presence ratings were higher when using head-controlled navigation, which allowed more fluid navigation strategies.

6. CONCLUSIONS AND FUTURE WORK

In this paper, we have presented a comparison between two 6DoF navigation techniques in virtual worlds. The first technique is a joystick-based technique. We chose it because this technique was similar to the ones used in 3D games for 6 DoF navigation tasks, or for piloting models of aircraft in the real world. For the second technique we focused on a specific head-controlled steering metaphor, not only because it is a hands-free navigation technique, but also because it allowed a vestibular stimulation of the user that we suspected might lead to reduced occurrence of cybersickness.

The conclusion of this study is that the head-controlled paradigm is clearly better than the joystick technique for 6DoF navigation in virtual spaces: it leads to improved user performance, a greater sensation of presence, and a lower occurrence of cybersickness.

Although these results are undoubtedly interesting for the design of future VR interfaces for tasks involving navigation in a 3D space, many questions remain as to why head-controlled navigation led to these positive outcomes. We have suggested that these results might be due to an increased physical and psychological involvement of the user in the virtual environment. Further studies will be required to verify this claim and to understand the connections between full-body involvement, presence, and the occurrence of cybersickness symptoms. As a first stage, we intend to study the use of head-controlled navigation in virtual scenes that offer fewer visual cues for the user to adapt his/her spatial orientation. We will also conduct studies to understand the impact of head-controlled navigation on cybersickness from a physiological point of view.

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